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Household energy consumption: a study of the role of occupants

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To what extent is energy performance determined by interactions between occupants, behaviour and buildings systems, as well as building and climate characteristics? The purpose of this paper is to establish which factors, and to what extent, affect the energy performance of a housing stock. The US residential sector is used to test the relative roles of socio-economic and behavioural aspects of occupants, as compared with climatic and physical building characteristics. Household heating and cooling energy consumption is investigated using an existing extensive survey, the Residential Energy Consumption Survey (RECS) by the US Department of Energy. This data set consists of information on actual energy consumption along with detailed energy-related characteristics of the housing units and their occupants. The paper presents a statistical analysis of the housing stock and assesses the relative roles of variables on energy use. Regression models and path analysis are used to determine the significance of parameters and their interrelationships. Results reveal that apart from climate, occupant behavioural aspects and socio-economic aspects are critical, partly directly and also significantly indirectly through their influence on choices and decisions about the physical characteristics of buildings and systems. Models are created for the analysis of heating and cooling energy that can be used to explore energy-related policies.

Keywords: behaviour, building stock, energy consumption, housing, inhabitant, residential, US

Dans quelle mesure la performance énergétique est-elle déterminée par les interactions entre les occupants, le comportement et les systèmes des bâtiments, ainsi que par les caractéristiques de la construction et du climat? L'objet de cet article est de déterminer les facteurs qui ont une incidence sur la performance énergétique d'un parc de logements, et ce, dans quelle mesure. Le secteur résidentiel américain est utilisé pour tester les rôles respectifs des aspects socioéconomiques et comportementaux des occupants, par comparaison avec les caractéristiques climatiques et physiques des bâtiments. La consommation énergétique domestique utilisée pour chauffer et refroidir les logements est étudiée en utilisant une vaste enquête existante du Ministère de l'Energie américain, l'Enquête sur la Consommation d'Energie du Secteur Résidentiel (RECS). Ce jeu de données consiste en des informations portant sur la consommation réelle d'énergie, ainsi que sur les caractéristiques détaillées, liées à l'énergie, des unités d'habitation et de leurs occupants. L'article présente une analyse statistique du parc de logements et évalue les rôles respectifs des variables sur l'utilisation d'énergie. Des modèles de régression et une analyse des pistes causales sont utilisés afin de déterminer l'importance des paramètres et leurs interrelations. Les résultats révèlent que, le climat mis à part, les aspects comportementaux et les aspects socioéconomiques des occupants sont cruciaux, pour partie directement, mais également indirectement, de manière significative, par leur influence sur les choix et les décisions touchant aux caractéristiques physiques des bâtiments et des systèmes. Il est créé des modèles permettant d'analyser l'énergie servant au chauffage et au refroidissement, qui peuvent être utilisés afin d'étudier les politiques énergétiques pouvant être menées.

Mots clés: comportement, parc bâti, consommation d'énergie, logement, habitant, résidentiel, États-Unis

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Introduction

The existing building stock is a major 'millstone' around our necks that holds the key to meeting national and international environmental targets for CO_2 emission reductions. Lomas (2009, p. 191) recently referred to this as 'one of the greatest challenges that nations currently face'. It is possible to demonstrate that new-build projects can achieve significant savings of 40–60% compared with existing buildings (*e.g.* Lowe, 2007; Boardman, 2007). However, new-build typically accounts for only 1–2% of the annual rate of change in the existing building stock (in developed countries), although some researchers propose that this rate should quadruple in order to meet emissions targets (Boardman, 2007). There is now a general agreement, amongst both scientists and policy-makers, that addressing the performance of the existing building stock has become a priority and a significant challenge.

Globally, 30–40% of all primary energy is used in buildings, and the residential sector accounts for the major part of this (United Nations Environment Programme (UNEP), 2007). The US has the largest CO_2 emissions per capita and the second largest in absolute terms in the world. In the US, residential buildings account for 21% of all the national CO_2 emissions, compared with 18% for commercial buildings which are a much more diverse and complex building stock (US Environmental Protection Agency (EPA), 2009). The US residential building stock is thus a significant and valuable focus for study.

The study of the residential building stock has a long history, and the use of the US Department of Energy's Residential Energy Consumption Survey (RECS), used in this paper and first conducted in 1978, has proved to be an invaluable resource for those interested in energy-related matters. Models have been developed to explore and project energy use trends either in the form of 'bottom-up' models using building physics to predict the effect of interventions (*e.g.* improved insulation), or as in this study, they take the form of 'top-down' studies that treat the sector as an energy sink and use historic data to understand or project trends forward (Swan and Ugursal, 2009). Data sets such as RECS are thus used either to calibrate bottom-up models or to explore particular socio-economic issues related to energy demand, such as the rebound effect of thermal improvements (Hsueh and Gerner, 1993); the health effects of increasing insulation (Levy *et al.*, 2003); ownership and eco-labelling of appliances (Fernandez, 2000; Sanchez *et al.*, 2008); life cycle effects of heating and cooling systems (Shah *et al.*, 2008); the effects of teleworking (Kitou and Horvath, 2003); econometric studies taking into account energy pricing (Mendelsohn, 2001); and numerous studies on the impacts of climate change on residential energy demand (Rosenthal *et al.*, 1995; Mendelsohn, 2003; Mansur *et al.*, 2005; Huang, 2006) including an excellent overview prepared for the US Climate Change Science Program (Wilbanks *et al.*, 2008). What the present paper addresses is to what extent building energy performance, often assumed to relate primarily to physical building characteristics, is determined by interactions between occupants, behaviour and systems, as well as building and climate characteristics. It does not aim to extrapolate or focus on a specific aspect but rather to understand and explain the relative roles that such variables play in domestic space-conditioning.

This paper exploits existing building stock data for US housing to gain insight into the key parameters related to energy use for heating and cooling. Traditionally, in the disciplines of architecture and building engineering, energy demand is addressed through individual building modelling based on the physical parameters of building form, fabric and systems. Simplistic and generalized assumptions are made about occupant behaviour in terms of occupancy patterns and internal gains. However, it is clear that there are a number of such non-technical parameters that confound the predictions, and this paper explores to what extent these are apparent at the level of the whole building stock:

> The management of building stocks cannot be improved without taking into account the interrelations between economic, physical, social and cultural aspects.
>
> (Kohler and Hassler, 2002, p. 232)

This paper is structured in five main sections, starting with a description of the data used for the study. This is followed by three analysis sections looking at direct, indirect and path analysis of variables that determine energy use. The final section presents predictive regression models for heating and cooling energy.

The data

This study uses the Residential Energy Consumption Survey (RECS) data, conducted in 2001 by the US Department of Energy (RECS, 2001), which was the most recent data set available at the time of undertaking the research for this paper. The data set consists of information about the actual total energy consumption and related characteristics of the dwellings and their occupants, collected from 4822 housing units in 50 states including the District of Columbia. Household characteristics were obtained from personal interviews with the householders and information related to energy consumption was collected from the energy suppliers. As a result, the data set contains details related to climate, building, occupant behaviour and socio-economic characteristics, and system, alongside the

energy consumption. Energy for heating and cooling, the largest proportion of the total energy demand, was prepared by RECS based on the analysis of the actual total energy use, fuel type and the climatic context.

In order to focus this investigation on energy use in fixed buildings for domestic use only, the data set was edited to exclude mobile homes, dwellings used for business and buildings whose systems heated or cooled other buildings. As a result, the data set was reduced to 3358 for heating energy and, excluding dwellings without air-conditioning, 2718 for cooling energy analysis.

Tables 1 and 2 outline the variables investigated in this paper, and are classified into climate, building, occupant, equipment and behaviour categories. Other parameters available from the RECS data, such as kitchen and other appliances, were not used because they do not have consistently significant relationships with energy consumption ($p > 0.001$ and based on F-tests for all variables). Heating and cooling degree-days to a reference temperature of 65°F (18.3°C) were used to represent climate (HDD65 and CDD65, respectively). It should be noted that heating and cooling energy provided in the RECS data is based on their regression analysis with respect to fuel type and heating and cooling degree-days. The purpose of this analysis is thus not primarily to confirm that there is a correlation but to identify the relative significance of the other physical and behavioural parameters that determine heating and cooling loads.

Building parameters describe the physical characteristics of the dwellings, including the total floor area, the number of windows, the year of construction, and the type of house (*i.e.* detached, attached or apartment). The average floor area of all dwellings is 212 m^2 (excluding unconditioned garage spaces), with approximately three-quarters being single-family dwellings and one-quarter being apartments. The type of heating and cooling system is classified in terms of being centralized or decentralized.

The occupant data include the size of household (*i.e.* number of occupants, on average 2.65), the total annual household income, and the age of the householder. The behavioural characteristics are given in terms of how occupants use their space-conditioning systems, including the number of rooms heated

Table 1 Variables analysed for space heating use

Variable		Range	
Climate	Heating degree-days	Mean = 4447, SD = 1958	
Building	Floor area (m^2)	Mean = 213, SD = 127	
	Number of windows	Mean = 3.82, SD = 0.93	
	Year of construction	1 = Before 1940	5 = 1970–1979
		2 = 1940–1949	6 = 1980–1989
		3 = 1950–1959	7 = 1990–1999
		4 = 1960–1969	8 = 2000
	Type of housing units	1 = Single-family detached	3 = Apartment in building with two to four units
		2 = Single-family attached	4 = Apartment in building with five or more units
Occupant	Number of household members	Mean = 2.65, SD = 1.47	
	Total annual income	1 = Less than US$5000	6 = US$30 000–US$39 999
		2 = US$5000–US$9999	7 = US$40 000–US$49 999
		3 = US$10 000–US$14 999	8 = US$50 000–US$74 999
		4 = US$15 000–US$19 999	9 = US$75 000–US$99 999
		5 = US$20 000–US$29 999	10 = US$100 000 or more
	Age of householder	Mean = 50.89, SD = 17.12	
Equipment	Type of space heating equipment	1 = Local systems including portable heaters and heating stove	
		2 = Local systems including built-in units and heat pump without ducts	
		3 = Central systems	
Behaviour	Number of heated rooms	Mean = 5.63, SD = 1.91	
		Mean = 20.01, SD = 2.50	
	Average temperature setting	13 = Lower than 13.5°C	19 = 18.5–19.49°C
		14 = 13.5–14.49°C	20 = 19.5–20.49°C
		15 = 14.5–15.49°C	21 = 20.5–21.49°C
		16 = 15.5–16.49°C	22 = 21.5–22.49°C
		17 = 16.5–17.49°C	23 = 22.5–23.49°C
		18 = 17.5–18.49°C	24 = Higher than 23.5°C
Energy	Total energy use for space heating (kWh/year)	Mean = 14 365, SD = 11 965	

Note: SD, standard deviation.

Table 2 Variables analysed for space cooling use

Variable		Range	
Climate	Cooling degree-days	Mean = 1470, SD = 926	
Building	Total floor area (m^2)	Mean = 209, SD = 128	
	Number of windows	Mean = 3.78, SD = 0.97	
	Year of construction	1 = Before 1940	5 = 1970–1979
		2 = 1940–1949	6 = 1980–1989
		3 = 1950–1959	7 = 1990–1999
		4 = 1960–1969	8 = 2000
	Type of housing units	1 = Single-family detached	3 = Apartment in building with two to four units
		2 = Single-family attached	4 = Apartment in building with 5 or more units
Occupant	Number of household members	Mean = 2.61, SD = 1.38	
	Total annual income	1 = Less than US$5000	6 = US$30 000–US$39 999
		2 = US$5000–US$9999	7 = US$40 000–US$49 999
		3 = US$10 000–US$14 999	8 = US$50 000–US$74 999
		4 = US$15 000–US$19 999	9 = US$75 000–US$99 999
		5 = US$20 000–US$29 999	10 = US$100 000 or more
	Age of householder	Mean = 51.45, SD = 16.76	
Equipment	Type of air-conditioning equipment	1 = Individual units	
		2 = Central system	
		3 = Both central and individual units	
Behaviour	Number of cooled rooms	Mean = 4.91, SD = 2.36	
	Frequency of air-conditioning use	0 = Not use at all	2 = Turned on quite a bit
		1 = Turned on only a few days or nights	3 = Turned on just about all summer
Energy	Total energy use for space cooling (kWh/year)	Mean = 2136, SD = 2104	

Note: SD, standard deviation.

(average 5.6) or cooled (average 4.9), and the thermostat settings for winter and summer (average 20°C). With respect to air-conditioning use, 38% of households that have it use it throughout the summer, 20% use it 'quite a bit', and 39% only use it 'when really needed'.

In order to carry out statistical comparisons between the parameters, it was checked whether all the data were normally distributed, where the data were graphically distributed in a symmetrical bell-shaped graph peaking at a mean, which is a requirement for such an analysis. Only the total floor area, cooling degree-days and space heating energy consumption did not meet the condition of normality, *i.e.* they had a skewness (a measure of asymmetry) and kurtosis (a measure of 'peakedness') of less than 1, where values of zero represented a perfect Gaussian distribution. For example, the heating energy data had a skewness of 1.615 and a kurtosis of 6.558. For these variables, a square-root transformation was applied to improve the normality.

Analysis of factors that directly affect energy use

Each variable is investigated using the General Linear Model (Johnston, 1978) to reveal the relationship between the individual variables and energy consumption. Results are shown in Tables 3 and 4. The *F*-statistic provides a measure of probability that energy use and the variable tested have the same variance, where the statistic is larger if the null hypothesis is not true. The *p*-value is a measure of the probability of obtaining a result at least as extreme as that actually observed, so the lower the value (usually below 0.05 or 0.01), the more significant the result (in this case all *p*-values are below 0.001). Finally, the coefficient of determination, R^2, represents the proportion of variability in the data that are accounted for by the variable and gives information about the goodness-of-fit. Values range from zero to 1, where 1 indicates a perfect fit, but in this context of social and behavioural factors values are lower than would be found in controlled physical experiments.

Climate and physical housing variables

It would be expected that climate is shown to have a strong influence on heating and cooling energy consumption given that these RECS energy data are prepared based on regression analysis of degree-days and total energy use. The correlation is furthermore not surprising for a housing stock that contains approximately 75% of buildings that predate energy-efficiency regulations. Heating degree-days is the single most significant variable influencing heating energy use

Table 3 General Linear Model (GLM) analysis for space heating energy consumption

Variables		Statistic	*p*-value	R^2
Weather	Heating degree-days	$F(1, 3495) = 1538.457$	<0.001	0.306
Building	Total floor area (m^2)	$F(1, 3495) = 963.265$	<0.001	0.216
	Number of windows	$F(6, 3490) = 136.630$	<0.001	0.190
	Year of construction	$F(7, 3489) = 71.512$	<0.001	0.125
	Type of housing unit	$F(3, 3493) = 171.900$	<0.001	0.129
Occupant	Number of household members	$F(10, 3486) = 3.967$	<0.001	0.011
	Total annual income	$F(9, 3487) = 14.206$	<0.001	0.035
	Age of householder	$F(1, 3424) = 47.096$	<0.001	0.014
Equipment	Type of heating	$F(2, 3473) = 438.343$	<0.001	0.202
Behaviour	Number of heated rooms	$F(14, 3461) = 54.803$	<0.001	0.181
	Average set temperature	$F(11, 3359) = 33.599$	<0.001	0.099

Table 4 General Linear Model (GLM) analysis for space cooling energy consumption

Variables		Statistic	*p*-value	R^2
Weather	Cooling degree-days	$F(1, 2716) = 1680.937$	<0.001	0.382
Building	Total floor area (m^2)	$F(1, 2716) = 234$	<0.001	0.079
	Number of windows	$F(6, 2711) = 25.921$	<0.001	0.054
	Year of construction	$F(7, 2709) = 30.779$	<0.001	0.074
	Type of housing unit	$F(3, 2714) = 82.970$	<0.001	0.084
Occupant	Number of household members	$F(10, 2707) = 22.163$	<0.001	0.076
	Total annual income	$F(9, 2708) = 27.080$	<0.001	0.083
	Age of householder	$F(1, 2659) = 66.586$	<0.001	0.024
Equipment	Type of cooling	$F(2, 2715) = 411.246$	<0.001	0.233
Behaviour	Number of cooled rooms	$F(14, 2703) = 147.639$	<0.001	0.433
	Use of air-conditioning	$F(3, 2714) = 789.516$	<0.001	0.466

($R^2 = 0.306$), but clearly does not sufficiently explain energy demand. For cooling energy the key variables were behavioural; in particular, the extent of use of air-conditioning ($R^2 = 0.466$) and the number of rooms air-conditioned ($R^2 = 0.433$); followed by climate ($R^2 = 0.382$). The type of system also has an important impact, such that households with local air-conditioning systems (*i.e.* individual units in walls or windows) consume only 46% of the cooling energy in households with centralized systems.

The physical characteristics of buildings affect the heating energy, but have a marginal effect on cooling energy. As the floor area of a dwelling increases so does the households' heating demand ($R^2 = 0.216$) to the extent that it is the second most important variable. However, for cooling energy the floor area ranks only seventh in importance of the eleven variables ($R^2 = 0.079$). The age of the building, and by implication its physical condition, has an inverse relationship with heating energy, such that houses built in the 1940s require 25% more heating energy than those built in the 1980s. As expected, detached houses on average consume twice as much heating energy as apartments in a building with five or more units (RECS, 2001).

Occupant choices and behaviour

Occupant behaviour is particularly significant for cooling energy in air-conditioned dwellings. The reported frequency of air-conditioning use accounts for 47% of the variation in cooling energy. The average cooling energy is for households that reported using air-conditioning occasionally, half that for households that use air-conditioning all summer. Behaviour has a less dramatic effect on heating energy than the physical characteristics of the climate, building and system. However, there is a direct relationship between the number of rooms heated by occupants and heating energy ($R^2 = 0.181$) and only a 10% variation in space-heating energy use was explained by the indoor thermostat setting chosen by householders. This latter point may be partially related to a poor awareness of temperature settings, although where possible these data were obtained from the actual thermostat setting.

Economic and demographic factors individually play a rather limited role in determining domestic energy use, although the size of the household, income and age of the householder are statistically significant, even if weak ($R^2 < 0.10$ for all three variables). During the winter this is explained by the fact that most houses are continuously centrally heated and thus socio-economic factors play a less significant role than climate, as confirmed by the analysis, but in summer the use of air-conditioning is more at the discretion of the occupant. One suspects that there are potential interactions between occupant variables that may in combination with other variables have an important effect on energy use. For example, household income is likely to influence the building type and choice of equipment. The next section will thus move on from assessing the direct effect of each individual parameter on energy use to explore the complex interactions in detail.

Analysis of indirect factors affecting energy use

We suggest that a more comprehensive analysis of household energy consumption requires the consideration of not only direct influences, but also indirect influences of the variables. The direct relationship between income and energy use is relatively weak. However, indirectly the consequences of increased wealth may be linked to a larger house, more air-conditioned and/or heated rooms, and centralized systems which in turn have been shown here to be significant variables.

Income

This section investigates the relationships between household income, on the one hand, and the physical and behavioural characteristics of households, on the other hand. The results are shown in Table 5. The three most important correlations with rising income are an increased number of heated rooms ($R^2 = 0.215$), an increased floor area of the house ($R^2 = 0.202$), and an increased number of air-conditioned rooms ($R^2 = 0.147$). Each of these, in turn, as previously shown, relate directly to energy use. For example, in the above analysis the direct links between occupant characteristics and cooling energy was weak ($R^2 < 0.10$). However, it is now known that increased income relates to increased air-conditioned rooms and that this in turn relates strongly to cooling energy use ($R^2 = 0.433$) (Table 4). Income does not influence comfort temperature settings.

Table 5 Influences of occupant economic status (*i.e.* income) on building, equipment, and behaviour factors

Variables	Statistic	*p*-value	R^2
Total floor area (m^2)	$F(9, 3487) = 97.776$	<0.001	0.202
Type of housing units	$F(9, 3487) = 34.439$	<0.001	0.082
Number of household members	$F(9, 3487) = 27.859$	<0.001	0.067
Number of heated rooms	$F(9, 3466) = 105.221$	<0.001	0.215
Average temperature	$F(9, 2975) = 1.719$	0.079	0.005
Number of cooled rooms	$F(9, 2708) = 51.693$	<0.001	0.147
Use of air-conditioning	$F(9, 2708) = 4.955$	<0.001	0.016

It is thus clear that an understanding of the energy consumption patterns requires the study of interrelationships between building, occupant, systems and behavioural aspects. This will enable one to identify the true indirect, as well as direct, effect of each aspect. This will be explored in the next section.

Path analysis

Path analysis identifies the causal relations between variables by analysing the relative strengths of the direct and indirect links between variables (Johnston, 1978). In order to explore these effects with respect to variables under consideration, path analysis is applied and the results discussed.

Space heating

Figure 1 is the path diagram for space heating energy demand, where the density of each path line represents the strength of the direct relationships between connected variables. Table 6 lists the values for the path lines, where the total effect of a variable is the sum of its direct and indirect effects. A value for a path line is a standardized partial regression coefficient from multiple regression analysis. Thus, for example, when there is an increase of 1 standard deviation (SD) in household income, the standardized household size increases by 0.466 for the heating analysis.

The physical aspects – such as the climate, heating type, age of house, and size of house – are the most significant determinants of household heating energy. However, they are closely related to occupant factors such as income and age of the householder. Income has the strongest indirect effect on heating energy of all the parameters, followed by age of the householder and number of occupants (despite there being no direct effect). Furthermore, income influenced every one of the physical building parameters, most significantly the floor area, number of windows and type of house. Occupants tend to lower the heating set temperature as the heating degree-days increase (*i.e.* in cold weather a lower indoor temperature is tolerated),

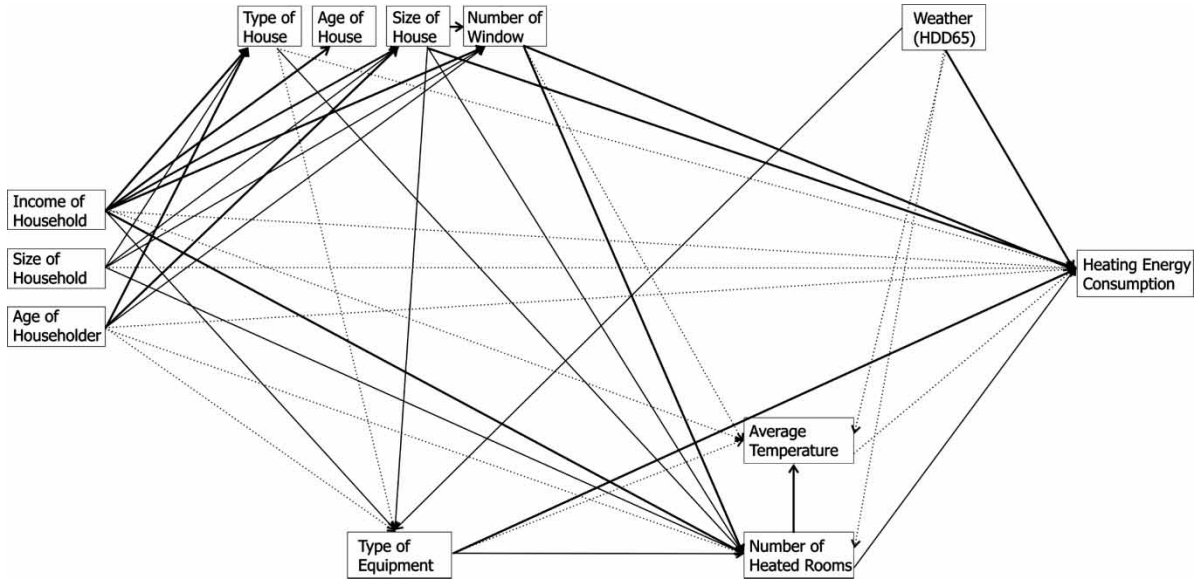


Figure 1 Path diagram for heating energy consumption

as shown by the positive effect (0.132) of heating degree-days on average set temperature in Table 6.

Space cooling

Cooling is most significantly related, after the climate, to the number of rooms that are air-conditioned and the way the systems are used (Figure 2 and Table 7). In other words, the air-conditioning energy use is significantly determined by occupant behaviour and choices, and less so by physical buildings characteristics. As with heating energy, the indirect effects through mediating factors such as the physical characteristics of houses, rather than direct effects, are dominant in cooling energy consumption. It is interesting to note that how occupants use their cooling equipment is only weakly related to household income. For example, the frequency of air-conditioning use is not statistically related to income, but use is related to climate and number of rooms cooled.

Models to predict heating and cooling energy

This section develops a predictive regression model for the square-root of heating and cooling energy consumption. The model enables one to predict energy use and assess the relative effects of variables on this energy use. The models are based on multiple regression analysis and are statistically and substantively significant ($R^2 = 0.640$ for heating and 0.817 for cooling energy). Tables 8 and 9 show the predictive models and related statistical tests: standard errors are small; *t*-tests (the ratio of the standard error to the value of the variable) are high; *p*-values are all lower than 0.05 confirming statistically significance; and tolerance values are all higher than 0.2 and thus confirm that the models are free from co-linearity problems (where two variables may be highly correlated and thus the model may not give valid results about an individual predictor).

Heating energy model

Table 8 shows the heating energy model and reveals that two parameters, namely the number of household members and age of the householder, do not have a directly significant effect. These have therefore been excluded from the model. The model can predict heating energy consumption of households as a function of climatic, physical, economic and behavioural factors.

The model can be used to explore, for example, the consequence of a 10% increase in heating degree-days to represent climate change. This has the effect of reducing heating energy by less than 0.1%. This is smaller than anticipated changes that might result from an economic recession with a reduction of household income of 10%, leading to a 1.6% reduction in heating energy use. Changes in occupant behaviour and expectation, such as increasing the set temperature by 1°C or extending the heating system to include another room (within the existing house), increase heating energy by 2.1% and 8.4%, respectively.

Cooling energy model

The model in Table 9 shows that not all the variables investigated have a significant relationship with cooling energy. The age of the property and number of windows do not have a directly significant effect

Table 6 Standardized total effects for heating energy consumption, also showing direct and indirect effects

	Heating degree-days	Total floor area (m^2)	Number of windows	Year of construction	Type of housing unit	Type of heating	Number of household members	Total annual income	Age of householder	Number of heated rooms	Average set temperature
Type of house	0.000	0.000	0.000	0.000	0.000	0.000	−0.176	−0.286	−0.275	0.000	0.000
Age of house	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.204	0.000	0.000	0.000
Floor area	0.000	0.000	0.000	0.000	0.000	0.000	0.159	0.466	0.234	0.000	0.000
Number of windows	0.000	0.000	0.000	0.000	0.000	0.000	0.155	0.335	0.170	0.000	0.000
Heating type	0.133	0.179	0.000	−0.058	0.034	0.000	0.022	0.177	0.032	0.000	0.000
Heated rooms	0.089	0.030	0.276	−0.010	−0.159	0.166	0.252	0.427	0.171	0.000	0.000
Average temperature	−0.132	0.003	−0.043	−0.001	−0.016	0.017	0.014	−0.052	0.005	0.101	0.000
Direct	0.459	0.198	0.151	−0.257	−0.098	0.298	0.000	−0.052	0.000	0.117	0.069
Indirect	0.041	0.057	0.029	−0.019	−0.009	0.021	0.109	0.218	0.129	0.007	0.000
Space heating	0.499	0.255	0.180	−0.276	−0.108	0.318	0.109	0.166	0.129	0.124	0.069

Note: Due to rounding, some data may not sum to totals.

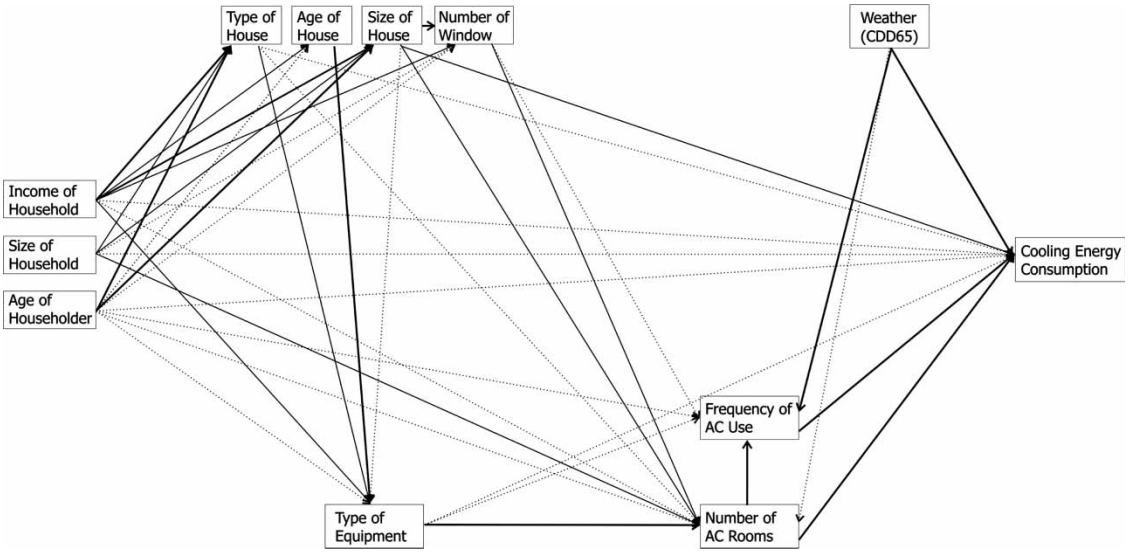


Figure 2 Path diagram for cooling energy consumption

and are thus excluded from the predictive model. The remaining variables in Table 9 are all significant as confirmed by the *t*-tests and a significance level of $p<0.05$, which implies that the estimation of results from the predictive model is reliable and robust.

The model can be used to explore a number of simple scenarios, for example, to predict the cooling energy implications of global warming, which would increase the cooling degree-days. The US Department of Energy (DOE) estimates that in 2010 the number of cooling degree-days will be 10% higher than the past 30 year average (DOE, Energy Information Administration (EIA), 2008). Applying this change in the model, and comparing it with assuming mean values of predictors for the whole building stock, shows an increase of 7.4% in cooling energy consumption of the stock. This value is of a similar order of magnitude to those produced by others (Wilbanks *et al.*, 2008) and represents the effect of climate change on the building stock in 2010.

Household income will often decrease by as much as 10% as a result of an economic recession. If this assumption is applied in the model, then the consequence is a reduction in cooling energy use of only 0.6% (assuming there are no further changes in behaviour such as the frequency of use or the number of rooms conditioned).

It is interesting to contrast the effect of climate change or income with, for example, a change in behaviour that increases the frequency of use of air-conditioning by one category (*e.g.* from 'turned on quite a bit' to 'turned on just about all summer'). This is a change, the scale of which is admittedly chosen arbitrarily, that could be brought about by both climate change and increased expectations of comfort. Such a behavioural change would result in an increased cooling load of 37.7%, demonstrating that changes in occupant behaviour potentially have very significant consequences on energy use.

It should be noted that in the application of the models, assumptions include that indoor set temperatures are fixed, the building stock remains unaltered, and there is no uptake of domestic air-conditioning or heating, unless stated in the analysis.

Discussion and conclusions

This paper has highlighted the fact that climate and building characteristics alone are insufficient as determinants of energy demand, and that the roles of occupant behaviour and socio-economic factors are important components. However, the most significant parameter that determines energy use is the climate. This suggests that global warming is likely to have a significant impact on the current building stock because it does not incorporate passive cooling strategies and occupants tend to control systems in close correlation to outdoor air temperatures (not indoor conditions). The second most important set of parameters relates to occupant behaviour, specifically in terms of the choices made about heating and cooling systems and their control. Less important than might be expected, perhaps surprisingly, are some physical characteristics of the dwellings (age, type and area), primarily because a large portion of the existing building stock was not designed to respond to meet current comfort standards through energy-efficient design. However, it is important to reiterate that houses from the 1940s perform better in terms of heating

Table 7 Standardized total effects for cooling energy consumption, also showing direct and indirect effects

	Cooling degree-days	Total floor area (m^2)	Number of windows	Year of construction	Type of housing unit	Type of cooling system	Number of household members	Total annual income	Age of householder	Number of cooled rooms	Use of air-conditioning
Type of house	0.000	0.000	0.000	0.000	0.000	0.000	−0.168	−0.304	−0.268	0.000	0.000
Age of house	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.181	−0.060	0.000	0.000
Floor area	0.000	0.000	0.000	0.000	0.000	0.000	0.151	0.473	0.214	0.000	0.000
Number of windows	0.000	0.479	0.000	0.000	0.000	0.000	0.152	0.339	0.183	0.000	0.000
Cooling type	0.000	0.041	0.000	0.395	−0.138	0.000	0.029	0.246	0.087	0.000	0.000
Cooled rooms	0.071	0.260	0.127	0.182	−0.152	0.460	0.192	0.364	0.161	0.000	0.000
Use of air-conditioning	0.375	0.070	0.006	0.077	−0.056	0.194	0.059	0.117	−0.034	0.326	0.000
Direct	0.455	0.137	0.000	0.000	−0.023	0.061	0.064	0.025	−0.086	0.341	0.334
Indirect	0.150	0.115	0.045	0.112	−0.079	0.222	0.112	0.251	0.085	0.109	0.000
Space cooling	0.605	0.252	0.045	0.112	−0.102	0.283	0.175	0.276	−0.001	0.450	0.334

Note: Due to rounding, some data may not sum to totals.

Table 8 Predictive regression mode for the square-root of heating energy consumption

Square-root of ($Y_{Heating}$) = $b_1X_1 + b_2X_2 + b_3X_3 + b_4X_4 + b_5X_5 + b_6X_6 + b_7X_7 + b_8X_8 + b_9X_9 + b_0$ $F(9, 3361) = 483.445$, $p < 0.001$, $R^2 = 0.640$

Coefficient	b_i	standard error (SE)	*t*-statistic	*p*-value	Partial correlation	Tolerance
X_1 – Average temperature	0.746	0.178	4.201	0.000	0.072	0.938
X_2 – Income	−0.942	0.258	−3.644	0.000	−0.063	0.698
X_3 – Type of heating	25.356	1.049	24.175	0.000	0.385	0.893
X_4 – Type of house	−4.068	0.641	−6.349	0.000	−0.109	0.679
X_5 – Number of windows	7.484	0.712	10.518	0.000	0.179	0.561
X_6 – Heated rooms	2.973	0.348	8.542	0.000	0.146	0.544
X_7 – Square-root of heating degree-days	0.010	0.000	37.071	0.000	0.539	0.920
X_8 – Square-root of floor area	2.092	0.173	12.119	0.000	0.205	0.527
X_9 – Year of construction	5.975	0.272	21.987	0.000	0.355	0.878
Constant	−105.804	5.100	−20.744	0.000		

Table 9 Predictive regression mode for the square-root of cooling energy consumption

Square-root of ($Y_{Cooling}$) = $b_1X_1 + b_2X_2 + b_3X_3 + b_4X_4 + b_5X_5 + b_6X_6 + b_7X_7 + b_8X_8 + b_9X_9 + b_0$ $F(9, 2651) = 1310.775$, $p < 0.001$, $R^2 = 0.817$

Coefficient	b_i	standard error (SE)	*t*-statistic	*p*-value	Partial correlation	Tolerance
X_1 – Use of air-conditioning	7.459	0.221	33.714	0.000	0.548	0.718
X_2 – Type of cooling	2.464	0.438	5.630	0.000	0.109	0.652
X_3 – Square-root of cooling degree-days	0.835	0.017	47.901	0.000	0.681	0.792
X_4 – Number of occupants	0.948	0.143	6.627	0.000	0.128	0.739
X_5 – Age of householder	−0.104	0.012	−8.673	0.000	−0.166	0.729
X_6 – Income	0.214	0.092	2.336	0.020	0.045	0.647
X_7 – Type of house	−0.427	0.197	−2.168	0.030	−0.042	0.718
X_8 – Square-root of floor area	0.654	0.054	12.123	0.000	0.229	0.595
X_9 – Number of rooms cooled	2.949	0.105	27.992	0.000	0.478	0.489
Constant	−29.502	1.521	−19.391	0.000		

demand than those from the 1980s. Those houses built before centralized heating or cooling – approximately 25% of the current stock – have by necessity adopted passive and climatic responses, although they may not meet today's generally expected combination of comfort and energy-efficiency standards. Another insight is that income is indirectly correlated with energy demand, via floor area and house form. These relationships have been quantified in detail herein and are summarized in Figure 3, which shows that energy use is directly affected by conventionally modelled parameters related to climate, building and equipment, but that these are in turn influencing behaviour and influenced by occupant socio-economic factors. The central role of occupant and behavioural aspects is evident and is essential for understanding, evaluating or predicting building energy use. Methodologies that do not take cognition of this are missing an essential component that determines energy demand.

Finally, the paper has presented models of heating and cooling energy that can be used to explore potential implications for energy use of the current building stock of policies or trends in relation to the physical and occupant-related variables. These might include increased cooling loads and reduced heating energy due to long-term climate change, or reduced energy consumption as a result of a shorter-term economic recession, but more significantly reflect the impact of behavioural changes of the occupants.

The next stage in the analysis will be to scrutinize cooling energy demand in greater detail. Although this is not the most significant component of the total energy demand – which is dominated by heating – air-conditioning energy demand responds strongly to issues of climate change. Currently, the findings suggest that an increase in cooling will outweigh the reduction in heating energy in the domestic sector, as confirmed by others (Huang, 2006). The direct impact of climate change will be through increased cooling energy to offset increases in outdoor temperatures – a straightforward physical relationship that has been demonstrated herein – but it is also likely to impact indirectly through an increased uptake in domestic air-conditioning. This trend is likely to be

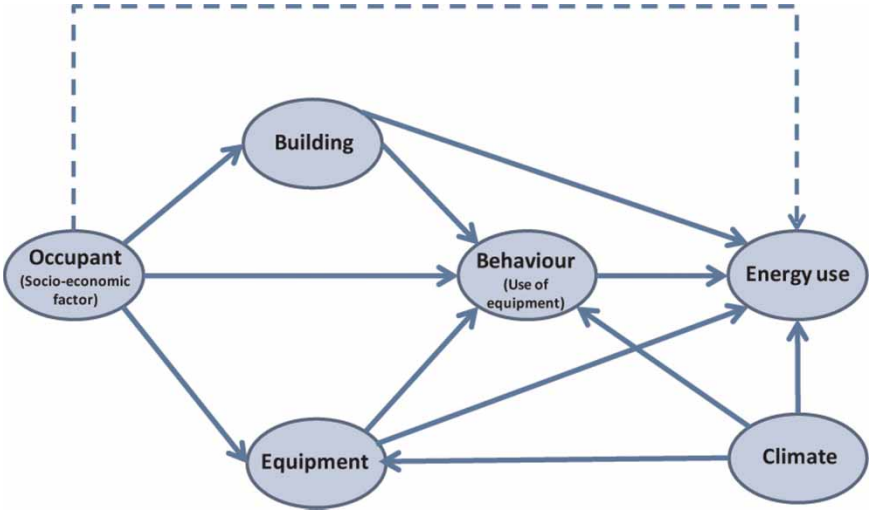


Figure 3 Diagrammatic representation of key-direct and -indirect links between physical (climate, building, equipment) and occupant (socio-economic, behavioural) factors that influence domestic energy use, and which should be considered in building stock modelling

reinforced by increased occupant expectations of comfort conditions, during increasingly hot spells, as well as an increase in floor space per household, projected to be 14% between 2003 and 2030 (US DOE, EIA, 2006). Strategies to counteract these trends in increased energy demand in existing buildings will need to include a study of the effects of physical interventions (such as shading, insulation, ventilation, controls, etc.) on occupant behaviour in order to be able to predict more confidently what the benefits will be across the whole building stock.

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